# Nathan Ludlow

Robotics Software Engineer

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I am a robotics software engineer who started in mechanical design and grew into full-stack autonomy. Over two years in industry I built perception, planning, and control software for industrial robot arms, including a domain-specific language that let LLMs program manipulators from natural language. Across four years of research I have published in reinforcement learning, autonomous driving, and physical human-robot collaboration. I want to bring effective robotics solutions to real-world problems.

## Technical Skills

**Programming**
Python, C++, C, Rust, Julia, MATLAB, Java

**ML / RL**
PyTorch, JAX / JaxMARL, TensorFlow, RLlib, Gymnasium, Weights & Biases; PPO, multi-agent & cooperative RL

**Robotics**
ROS / ROS2, Gazebo, MuJoCo, PyBullet, OpenCV, RealSense SDK

**Methods**
Perception (Kalman/EKF, point-cloud processing, pose estimation); planning (RRT, A*, MPPI, trajectory optimization); control (MPC, control barrier functions, LQR, impedance control); knowledge-graph reasoning, LLM integration

**Robot Platforms**
Yaskawa, xArm, Universal Robots, Franka Panda, KUKA, Farm-ng

**Hardware / CAD**
SolidWorks (Certified), Onshape, Inventor, strain-gauge sensing, embedded control

**Tools**
Linux, Git, Docker, CUDA, LaTeX

**Languages**
English (Native), Spanish (Fluent)

## Education

**Carnegie Mellon University**
M.S. in Robotics
*Advised by Dr. Katia Sycara — GPA 4.00/4.00*
Expected Jul. 2026 • Pittsburgh, PA

**Brigham Young University**
B.S. in Mechanical Engineering
*Emphasis: Dynamics, Controls, Robotics*
Minors: CS, Math — GPA 3.79/4.00
Apr. 2024 • Provo, UT

## Industry Experience

**Altitude AI** May 2022 – Sep. 2023
*Robotics Software Engineer | Salt Lake City, UT*

- **Automated meat processing:** Led a team to build the perception, path-planning, and control pipeline for robotic pork-belly skin removal, coordinating industrial arms with electric cutting tools on an assembly line.
- **Natural-language robot programming:** Designed a domain-specific language that let LLMs generate executable code for industrial arms from natural-language instructions, enabling customers to rapidly create and modify custom tasks on our hardware platforms and accelerating manipulation-task prototyping.
- Part of the team that developed and deployed an adaptive pick-and-place robotic system for meat processing at a customer facility, and ran on-site deployment tests of the pork-skinning system to validate performance under real production conditions.

## Research Experience

**CMU Advanced Agent Robotics (AART) Lab** Sep. 2024 – Present
*Graduate Research Assistant — Dr. Katia Sycara | Pittsburgh, PA*

- Developed Knowledge Graph-Augmented Reinforcement Learning (KG-RL), leveraging semantic relationships for cooperative multi-agent tasks; demonstrated up to **3× faster convergence** and improved final performance across Overcooked-AI, MiniGrid, Craftax, and AI-Habitat. *First-author submission under review.*
- Extending KG-RL to temporal knowledge graphs with attention-based and recurrent encoders (Time2Vec, Time-LSTM) to reason over dynamic state evolution and temporal dependencies.
- Engineered a vectorized JAX / JaxMARL training stack on lab GPU servers, building reproducible PPO training and evaluation pipelines for multi-agent benchmarks.

**CMU Driverless Intelligent Vehicles (DRIVE) Lab** Jun. 2023 – Aug. 2024
*RISS Research Fellow & Research Assistant — Dr. John M. Dolan | Pittsburgh, PA*

- Designed a hierarchical risk-aware behavioral model for simulating human drivers in AV testing, introducing a forward-looking Gaussian risk metric with social-pooling trajectory prediction. *First-author publication at ICRA 2024.*
- Developed a risk-aware Voronoi-cell approach inspired by control barrier functions for safe multi-agent navigation around dynamic obstacles under motion uncertainty.

**BYU Robotics And Dynamics (RAD) Laboratory** Feb. 2022 – Jun. 2024
*Research Assistant — Dr. Marc Killpack & Dr. John Salmon | Provo, UT*

- Developed dynamic control methods for physical human-robot co-manipulation of large objects (cooperative swinging, throwing, tossing) and built the full hardware/software architecture for an omnidirectional mobile base. *UCUR 2023.*
- Built a VR platform with force sensing to analyze human-human interaction dynamics during cooperative manipulation.

**BYU Crop Biomechanics Laboratory** Sep. 2023 – Apr. 2024
*Capstone Team Lead — Dr. Douglas Cook | Provo, UT*

- Led a team building an autonomous robotic system to measure maize-stalk stiffness in the field, designing a novel strain-gauge sensor package, processing algorithms, and navigation control. *Publication in MDPI Sensors 2025.*

## Publications


- **N. Ludlow**, V. Tadiparthi, H. N. Mahjoub, Y. Xie, W. Kim, K. Sycara, S. Stepputtis. “Knowledge Graph-Augmented Reinforcement Learning.” *TMLR*, 2026. *Under review.*
- **N. Ludlow**, A. Krieger, M. Menon, J. Chandrachud. “Strategy Search for Human-Robot Teaming.” *CHAI Workshop*, UC Berkeley, 2026. *To appear.*
- C. Noh, K. Smith, . . . , **N. Ludlow**, . . . , D. Cook. “Measurement of Force and Position Using a Cantilever Beam and Multiple Strain Gauges.” *MDPI Sensors*, 2025.
- **N. Ludlow**, Y. Lyu, J. M. Dolan. “Hierarchical Learned Risk-Aware Planning Framework for Human Driving Modeling.” *ICRA*, 2024.
- **N. Ludlow**, M. D. Killpack, J. L. Salmon. “Mobile Base for Physical Human-Robot Interaction and Co-manipulation.” *UCUR*, 2023.